The spiral model is a risk-driven software development process that combines the waterfall model's systematic steps with the repetitive cycles of iterative development. [1, 2]

Introduced by Barry Boehm in 1986, this approach helps teams build complex projects by repeating the same four core phases in expanding loops (or spirals). [1, 2, 3]

The Four Phases of a Spiral Cycle

Every loop in the spiral represents a single pass through four fundamental quadrants or phases: [1, 2]

- **Planning (Objective Setting):** Identify project goals, functional requirements, and alternative ways to meet them while noting any project constraints. [1, 2]
- **Risk Analysis:** Assess the alternatives to spot potential threats or uncertainties, often building prototypes to test and resolve high-risk items early. [1, 2]
- **Engineering (Development and Test):** Build, code, and test the actual features or prototype planned for the current cycle. [1, 2]
- **Evaluation (Review):** Review progress with stakeholders, gather customer feedback, and plan the next loop around the spiral. [1, 2]

Advantages and Disadvantages

Pros

- **Strong Risk Management:** Focuses heavily on identifying and fixing hazards before they ruin a project.
- **Handles Changing Requirements:** Allows teams to adapt scope and adjust features in later loops.
- **Early Feedback:** Users test early versions, which boosts overall satisfaction. [1, 2, 3, 4]

Cons

- **Complex to Run:** Requires skilled professionals who know how to analyze risks properly.
- **Expensive:** Takes a lot of time and resources, making it a poor choice for small or simple projects.
- **Unclear Schedules:** The number of loops is not fixed ahead of time, which makes time tracking hard. [1, 2, 3]

Here is a short video explaining the core concepts and phases of the spiral model in software engineering:



https://www.youtube.com/watch?v=VcanEac3xeg